MILITARY SPECIFICATION

INSULATION, MOLDED, NOISE-BARRIER

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a molded, expandable, self-skinning material used as a noise-barrier insulation (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking for US Military Property.
MIL-STD-45662	- Calibration System Requirements.

2.1.2 Other Government document. The following Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

NATIONAL HIGHWAY TRAFFIC SAFETY ADMINISTRATION

FMVSS 302	- Flammability of Interior Materials - Passenger Cars, Multipurpose Passenger Vehicles, Trucks, and Buses.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D412	- Rubber Properties in Tension.
ASTM D624	- Rubber Property - Tear Resistance.
ASTM E90	- Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA, 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish noise barrier material, herein referred to as insulation, which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information, shall be a sample of the units to be furnished to the Government. All subsequent insulation delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein, and shall be free from defects which would adversely affect the performance or serviceability of the finished product (see 4.8.1).

3.2.1 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials; however, all materials shall be new and unused. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.3.1).

3.3 Design and construction.

3.3.1 Configuration. The configuration of the insulation shall be as specified on the applicable drawing (see 4.8.2 and 6.2).

3.3.2 Color. The color of the insulation shall be black (see 4.8.2).

3.4 Physical properties.

3.4.1 Surface density. The surface density of 1-inch thick insulation shall be 1.2 pounds per square foot $\pm$ 10 percent (see 4.8.3).

3.4.2 Tear resistance. The tear resistance of the insulation shall be not less than 4 pounds force per inch (ppi) (see 4.8.4).

3.4.3 Tensile strength. The tensile strength of the insulation shall be not less than 25 pounds force per square inch (psi) (see 4.8.5).

3.4.4 Elongation. The elongation of the insulation shall be not less than 80 percent (see 4.8.6).

3.4.5 Flammability. The burn rate of the insulation shall be not greater than 2 inches per minute (see 4.8.7).

3.4.6 Impact resistance. The insulation at -50 degrees Fahrenheit (°F) shall be capable of withstanding an impact of not less than 15 centimeter-kilograms without cracking (see 4.8.8).

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3.4.7 Flexibility. The insulation shall be capable of passing a 1-inch mandrel bend without cracking after being exposed for 4 hours at 250°F and 4 hours at -50°F (see 4.8.9).

3.5 Acoustical performance. The acoustical performance of 1-inch thick insulation mounted on 20-gauge steel shall be as specified in table II (see 4.9).

TABLE I. Noise barrier properties.

Frequency (hertz)	Sound transmission loss (decibels)
160	16
250	20
500	21
1000	36
2000	50
4000	58

3.6 Identification marking. Identification marking shall be in accordance with MIL-STD-130 (see 4.8.2).

3.7 Workmanship. Workmanship shall be such that the insulation is uniform in color, mechanical properties, and density, and free of splay marks, excessive flash, porosity, checks, cracks, crazing, or other molding defects which would affect its life, serviceability, or appearance (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of

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assuring that all products or supplies submitted to the Government for acceptance comply wih all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
    1. Examination (see 4.5.2).
    2. Tests (see 4.5.3).
c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 $\pm$ 18°F
b. Barometric pressure 28.5 $^{+2}_{-3}$ inches mercury (Hg)
c. Relative humidity 50 $\pm$ 30 percent

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select insulation samples produced under the production contract (see 6.2) for first article inspection. First article samples shall be inspected as specified in table II. Approval of first article samples by the Government shall not relieve the contractor of his obligation to supply insulation that is fully representative of those inspected as first article samples. Any changes or deviation of the production units from the first article samples shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Test item deficiencies during, or as a result of, first article inspection shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, first article inspection shall be evidence that all items already produced prior to completion of first article inspection are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor at no additional cost to the Government. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

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TABLE II. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Tests	
Materials	3.2	4.8.1	X		X	
Defects (see 4.5.2 and table IV)	3.3.1, 3.3.2, 3.6 and 3.7	4.8.2	X	X		
						X
Surface density	3.4.1	4.8.3	X			X
Tear resistance	3.4.2	4.8.4	X			X
Tensile strength	3.4.3	4.8.5	X			X
Elongation	3.4.4	4.8.6	X			X
Flammability	3.4.5	4.8.7	X			X
Impact resistance	3.4.6	4.8.8	X			X
Flexibility	3.4.7	4.8.9	X			
Acoustical properties	3.5	4.9	X			X

4.5 Quality conformance inspections.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the insulation of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table III.

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TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Dimensions affecting interchangeability, out of tolerance (see 3.3.1).	SIE 1/
102	Identification marking, improper (see 3.6).	Visual
103	Faulty workmanship affecting performance (see 3.7).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3.1).	SIE
202	Color, improper (see 3.3.2).	Visual
203	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Tests. Each insulation sample shall be subjected to the quality conformance tests specified in table II.

4.6 Control tests. Unless otherwise specified (see 6.2), control tests shall be conducted on one insulation sample produced from each batch of new raw material. The insulation shall be subjected to the tests specified in table II.

4.7 Failure. Failure of any sample to pass any of the specified quality conformance or control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials. Conformance to 3.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

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4.8.2 Defects. Conformance to 3.3.1, 3.3.2, 3.6, and 3.7 shall be determined by examination for the defects listed in table III. All parts supplied to this specification must be allowed to cure at room ambient conditions for not less than 96 hours prior to checking dimensions, physical properties (except 4.8.8 and 4.8.9), or acoustical properties.

4.8.3 Surface density. To determine conformance to 3.4.1, the weight of an insulation sample shall be divided by its area.

4.8.4 Tear resistance. To determine conformance to 3.4.2, the insulation shall be tested as specified in ASTM D624, die C.

4.8.5 Tensile strength. To determine conformance to 3.4.3, the insulation shall be heat aged at 250 ± 10°F for 48 hours and tested as specified in ASTM D412, die C.

4.8.6 Elongation. To determine conformance to 3.4.4, the insulation shall be heat aged at 250 ± 10°F for 48 hours and tested as specified in ASTM D412, die C.

4.8.7 Flammability. To determine conformance to 3.4.5, the insulation shall be tested as specified in FMVSS 302.

4.8.8 Impact resistance. To determine conformance to 3.4.6, not less than five insulation samples 2 inches by 4 inches by 1-inch thick shall be conditioned at 77 ± 5°F and 50 ± 5% relative humidity for 7 days and then conditioned at -50 ± 2°F for not less than 4 hours. Using a Gardner impact tester having a 1-kilogram dart with a 1.6-centimeter (cm) diameter, the conditioned insulation samples shall be centered under the dart, and a steel plate (1 inch by 3 inches by 1/4-inch thick with a 1/8-inch radius on all edges) shall be centered on the conditioned insulation samples. The dart shall be raised to a height of 15 cm and dropped onto the steel plate. The sample shall be allowed to warm to room temperature for 15 minutes. The tested area shall then be bent around a 2-inch radius and examined for evidence of cracks on the top surface.

4.8.9 Flexibility. To determine conformance to 3.4.7, four insulation samples 1 inch by 8 inches by 1-inch thick shall be conditioned at 77 ± 5°F and 50 ± 5% relative humidity for 7 days; 250°F for 4 hours with a 1-hour recovery at 77 ± 5°F; and -50°F for 4 hours with a 1-hour recovery at 77 ± 5°F. The conditioned insulation samples shall then be bent around a 1-inch mandrel and examined for evidence of cracks on the top surface.

4.9 Acoustical performance. To determine conformance to 3.5, the insulation shall be tested as specified in ASTM E90.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

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6. NOTES

6.1 Intended use. The insulation covered by this specification is intended for use in applications such as tunnels, individual engine components, and any other application where a chemical-resistant, totally sealed, molded noise barrier is required.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If first article samples are not required (see 3.1).
c. Applicable drawing number, title and date (see 3.3.1).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
f. If inspection conditions shall be other than as specified (see 4.3).
g. If first article inspection is not required (see 4.4).
h. Number of samples to be selected for first article inspection (see 4.4).
i. If control testing shall be other than as specified (see 4.6).
j. Applicable levels of preservation, packaging, packing, and marking (see 5.1).

6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.2).

6.3.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

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6.4 Subject term (key word) listing.

Insulation
Molded insulation
Noise barrier

6.5 Supersession data. This military specification supersedes AM General specification AMG 191, revision A, dated 16 May 1984.

Custodian:
Army - AT

Preparing activity:
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(Project 5640-A443)

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